

Eli Foster

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EDUCATION

Bucknell University

Bachelor of Science in Electrical Engineering. Concentration in Semiconductor Technology.

GPA: 3.65 GPA / 3.77 Engineering GPA.

Lewisburg, PA

May 2027

EXPERIENCE

Electrical Engineering Intern, Battery Management Systems

June 2026 – August 2026

EOS Energy

- Supported Electronics R&D team by creating engineering documentation, verifying new designs, and testing and debugging prototypes
- Contributed to schematic capture and PCB layout for battery management systems operating up to several kilovolts, requiring precise design practices
- Assisted in bringing board designs from ideation through manufacturing, supporting large-scale production readiness
- Conducted validation and verification of prototype designs using benchtop equipment including oscilloscopes and power supplies; produced documentation supporting CE certification
- Followed Agile development methodology using Jira

Maker-E Project Developer

May 2024 – August 2025

Bucknell University

Lewisburg, PA

- Expanded and optimized the PortalBox embedded hardware platform to improve makerspace equipment access, focusing on analog and digital circuit layout and PCB design
- Conducted firmware testing, bug identification, and feature enhancements in collaboration with faculty and technical IT staff
- Created educational technology kits introducing students to microcontrollers, PCB design, computer architecture, and manufacturing equipment

Makerspace Manager

August 2024 – Present

Bucknell University

Lewisburg, PA

- Maintain and repair 3D printers, laser cutters, PCB assembly machines, and lab equipment
- Certify users on manufacturing equipment and provide training on best practices for electronics fabrication and testing
- Assist users with EE concepts including circuit design and layout, Python, Assembly, and C; serve as TA for several ECE courses

Bank Teller

May 2023 – August 2024

Newfield National Bank

Glassboro, NJ

- Customer service representative for several bank locations
- Responsible for handling money and maintaining a balanced drawer
- Fingerprints on file and a security clearance gained

PROJECTS

Find more on my portfolio website, www.foster.engineer

RFID Makerspace Equipment Access (PortalBox v5.0)

Summer 2025

- Completed v5.0 of the open-source PortalBox project, delivering an ESP32-S3-based interlock system for makerspace equipment
- Designed hardware using KiCAD and Altium Designer; implemented firmware in MicroPython, based on prior v4.0 Python firmware
- Verified user access against a hosted SQL database; used Git for version control and Embedded Linux to build and flash firmware
- Tested and debugged populated PCB using oscilloscopes, multimeters, and microscopes to ensure full functionality

Bucknell Baja SAE

August 2024 – May 2025

- Led a team of approximately 10 ECE engineers to design and implement all electrical systems on an off-road racing vehicle
- Served as client for an ECE Senior Design team, supplying requirements, specifications, and feedback throughout the design cycle
- Implemented a wiring harness, electronic differential controls, and an instrumentation system for vehicle monitoring
- Applied Agile-style development practices and produced technical documentation throughout the project

Microscope Manipulator

January 2026 – December 2026

- Contracted by Biology professor to improve an aging micromanipulator setup as commercially available solutions are insufficient
- Moved Raspberry Pi and complex wiring harness to one small, USB-controlled board with a Python GUI
- Designed custom PCB using FTDI MPSSE chip to translate USB into SPI for piezoelectric stage control
- Standalone power module integrated into a small section of the board, using a buck converter and a switch between voltage dividers for different voltage levels available